



Helix Learns to Fold Laundry

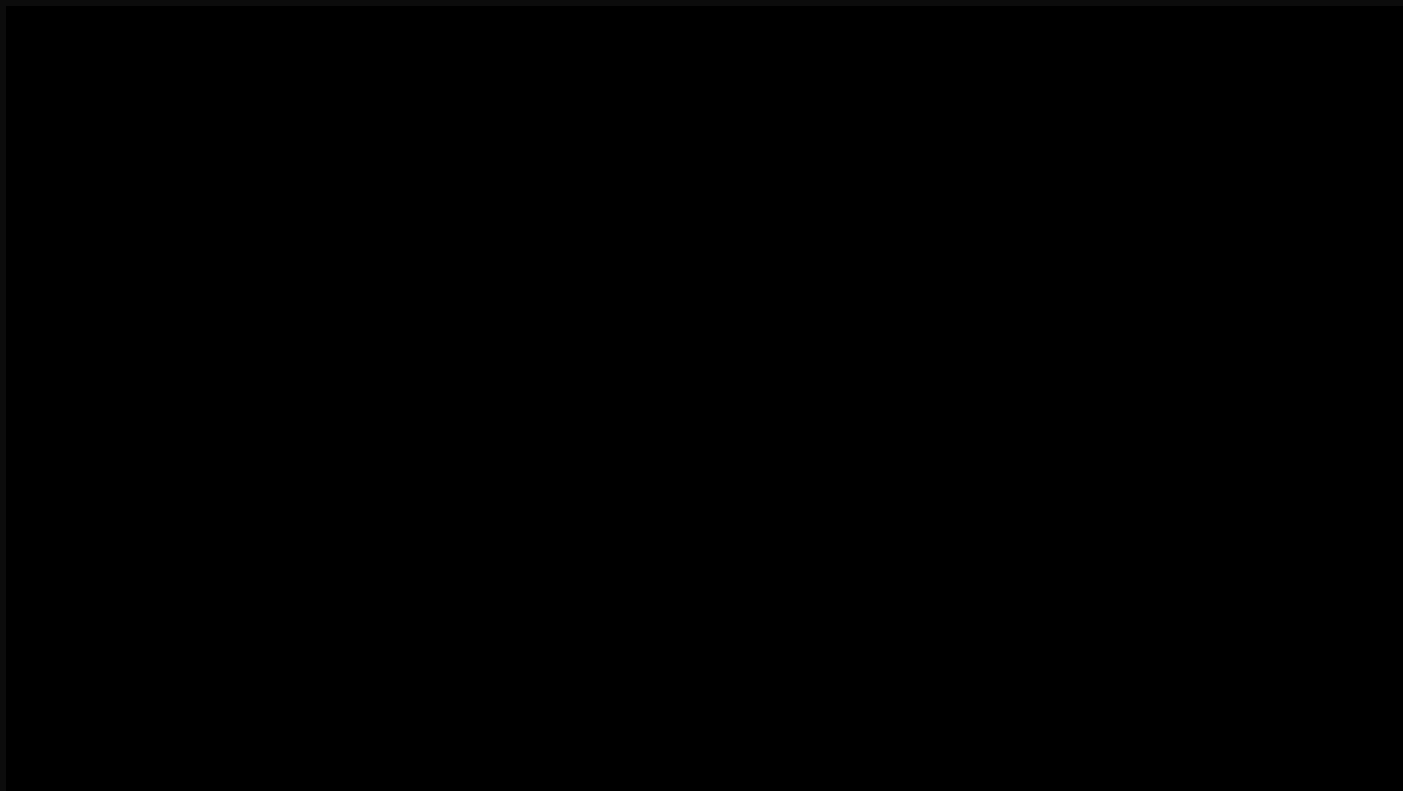
August 12, 2025

Helix, Figure's [Vision Language Action \(VLA\) model](#), recently demonstrated an hour of fully autonomous package reorientation in a logistics setting. Now, the same model is tackling something entirely different: folding laundry.

Folding laundry sounds mundane for a person, but this is one of the most challenging dexterous manipulation tasks for a humanoid robot. Towels are deformable, constantly changing shape, bending unpredictably, and prone to wrinkling or tangling. There's no fixed geometry to memorize, and no single "correct" grasp point. Even a slight slip of a finger can cause the material to bunch or fall. Success requires more than just seeing the world accurately - it demands fine, coordinated finger control to trace edges, pinch corners, smooth surfaces, and adapt in real time.

Key Results:

- **A first for humanoids.** This is the first instance of a humanoid robot with multi-fingered hands folding laundry fully autonomously using an end-to-end neural network.
- **Same architecture, data-only change.** The same Helix architecture that solved logistic tasks was applied directly to laundry folding - with no modifications to the model or training hyperparameters. The only addition was the dataset.
- **Natural multimodal interaction.** In addition to folding, Helix learned to maintain eye contact, direct its gaze, and use learned hand gestures while engaging with people.



Video 1: Helix folds towels autonomously.

Without any architectural changes, Helix learned to:

- Pick towels from a mixed pile.
- Adjust folding strategies based on starting configurations.
- Recover from multi-pick errors by returning extra items.
- Use fine manipulation skills, like tracing an edge with a thumb, pinching corners, or unraveling tangled towels - before completing folds.

Critically, Helix does all of this without explicit object-level representations. For highly deformable items like towels, building such representations is brittle and unreliable. Instead, Helix operates entirely end-to-end: from vision and language input to smooth, precise motor control.

Why this matters



The same general-purpose architecture, and the same physical platform, can seamlessly transition from industrial logistics to household chores. As we scale real-world data collection, we expect Helix's dexterity, speed, and generalization to keep improving across an even broader range of tasks.

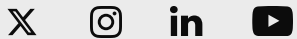
If you're interested in helping us push the frontier of general-purpose humanoid intelligence, [we're hiring](#).

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